Multi-Task Learning for OSA Detection and Sleep Staging via Multi-Scale Modeling

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***Abstract*—Obstructive sleep apnea (OSA) and sleep fragmentation are closely linked physiological phenomena that play crucial roles in the diagnosis and management of sleep disorders. While numerous deep learning models have been developed for either OSA detection or sleep stage classification, few attempts have been made to address both tasks simultaneously. To this end, we propose MT-TASPPNet (Multi-Task Triple Atrous Spatial Pyramid Pooling Network), a unified multi-modal multi-task network that jointly performs automatic OSA event detection and sleep staging. The model integrates modality-specific feature extractors for EEG, ECG, and airflow signals, and employs Atrous Spatial Pyramid Pooling modules in both the modality-specific and shared representation pathways to capture multi-scale temporal–frequency patterns. Additionally, an EOG-guided prior mechanism is incorporated to enhance the discrimination of subtle sleep stages. We use a 3-min input window (1-min target with $\pm$ 1-min context) and evaluate our method on three large-scale datasets: SHHS1, SHHS2, and Sydney Sleep Biobank. The model achieves OSA detection accuracy between 0.798 and 0.884 (MF1: 0.772 to 0.821), and sleep staging accuracy between 0.776 and 0.834 (MF1: 0.735 to 0.749, $\mathcal{K}$: 0.697 to 0.77). Notably, the model maintains consistent performance despite data heterogeneity and individual variability. These results validate the stability and adaptability of MT-TASPPNet in clinical settings, paving the way for efficient and scalable multi-task sleep analysis systems.**

***Index Terms*—Sleep monitoring, multi-scale model, multi-task learning, obstructive sleep apnea detection, sleep staging.**

I. Introduction

DESPITE the continuous expansion of research and clinical practice in sleep medicine, the current awareness of sleep disorders among the general public and healthcare community is still insufficient, severely limiting the establishment of early diagnosis and intervention systems [1]. Sleep disorders have become an important public health problem that needs to be solved globally, and chronic insomnia and sleep deprivation are closely related to the elevated risk of various chronic diseases such as hypertension, diabetes, obesity, depression and cardiovascular diseases [2], [3], [4]. According to the diagnostic criteria published by the American Academy of Sleep Medicine (AASM) [5], there is currently a lack of a uniform objective diagnostic basis for most sleep disorders, and patients often have difficulty in accurately describing their own symptoms, which increases the difficulty of clinical assessment. The diagnosis of sleep breathing disorders such as Obstructive Sleep Apnea (OSA) usually relies on subjective questionnaires and scales, combined with polysomnography (PSG), which is widely recognized as the "gold standard", and which usually consists of electroencephalogram (EEG), electrooculogram (EOG), electromyogram (EMG), electrocardiogram (ECG), oro-nasal airway, and other tests [6]. These physiological signals from PSG can be used to identify sleep architecture and respiratory events, and is the core basis for determining the type and severity of the disease. However, PSG data is complex, and the manual labeling process is time-consuming. Even an experienced technician needs 2 hours to complete the labeling of a night's record, which seriously restricts the development of large-scale screening and efficient diagnosis and treatment. Meanwhile, the rise of deep learning provides an effective path for automated PSG analysis, and end-to-end modeling is expected to ensure accuracy while significantly reducing labor costs, promoting intelligent assisted diagnosis of sleep disorders.

In most existing machine learning studies, sleep staging and OSA detection are treated as separate tasks, with EEG commonly used for staging and ECG or airflow signals for OSA detection [7], [8], [9], [16], [17], [18], [19], [20], [21]. However, existing studies and clinical experiences have shown that the occurrence of OSA events is closely related to sleep architecture, especially in Rapid Eye Movement (REM) stage, where reduced airway muscle tone is more likely to lead to airway collapse [10], [33]. At the same time, frequent OSA events can also interfere with normal sleep architecture, inducing arousal responses and stage transitions [11]. This physiological bidirectional correlation suggests that only modeling the two tasks independently cannot fully explore their potential synergistic relationship.

In this context, multi-task learning (MTL) offers a promising approach for jointly modeling sleep staging and OSA detection.

Received 12 June 2025; revised 26 September 2025 and 6 December 2025; accepted 17 December 2025. Date of publication 23 December 2025; date of current version 6 July 2026. This work was supported by the University of Sydney Nano Institute Kickstarter Grant, under Project 228599. *(Corresponding author: Wei Chen.)*

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Digital Object Identifier 10.1109/JBHI.2025.3647317

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MTL not only enhances model generalizability by constructing a shared semantic representation, but also improves the characterization of critical events through inter-task collaboration. Although a few studies have attempted to integrate both tasks within a unified framework, several challenges remain: (1) the fusion strategies for multi-modal signals are often coarse and fail to fully exploit complementary information; (2) the shared mechanisms between tasks lack structural constraints, potentially leading to semantic interference; and (3) the classification performance for key stages such as REM and N1 still requires further improvement.

To this end, this paper proposes a multi-modal multi-task collaborative network, MT-TASPPNet, to simultaneously realize the automatic detection of sleep staging and OSA events. The method has the following innovations:

- A parallel multi-modal feature extraction sub-network is constructed to process EEG, ECG and airflow signals separately to improve the modal characterization ability.
- An EOG guidance mechanism based on a priori statistical features is introduced to optimize the confusing discrimination between REM and N1 in sleep stages.
- Task-specific ASPP-based feature encoding is integrated with a cross-modal shared representation fusion layer to enable task decoupling and feature-level collaboration.
- Cosine consistency and contrastive learning regularization are incorporated to align inter-modal feature directions and enhance multi-task synergy efficiency.

II. Related Works

A. OSA Detection and Sleep Staging

Deep learning has been widely applied to the automatic detection of OSA and sleep staging based on physiological signals. Among these methods, the choice of input segment length has a significant impact on model performance. While many studies follow the 30-s standard in sleep physiology, some have explored longer contextual windows to enhance the modeling of temporal dependencies.

In the field of OSA detection, early studies focused on architectures such as one-dimensional convolutional neural network (CNN), recurrent neural network (RNN), and long short-term memory (LSTM) networks [7], [8], [9]. Akahin et al. [12] extracted EEG synchronization features from 30-s Stage 2 sleep segments to distinguish OSA, central sleep apnea (CSA), and controls with neural networks. Zhao et al. [13] classified 30-s EEG epochs into OSA, CSA, and normal events using sub-band entropy and variance features. Le et al. [14] partitioned the breathing-sound recordings into 30-s epochs and labeled each as "apnea," "hypopnea," or "no event," achieving about 86% accuracy.

As multiple studies [15], [16], [17] have shown that incorporating information from adjacent time segments can improve the accuracy of OSA detection for each segment, most works tend to use the current time segment along with several minutes of preceding and following signal segments as the model input, while assigning the event type of the current 60-s segment as the corresponding label, thereby covering the full temporal span of a single apnea event. DeepApneaNet [15] is a cascaded CNN-BiLSTM framework proposed for detecting OSA from single-lead ECG signals, achieving high accuracy without handcrafted features, and showing strong potential as a practical solution for real-world OSA screening. Wang et al. [16] developed a lightweight multi-scale network that takes a 5-minute ECG segment as input and uses the event category of the middle 1-minute segment as the label, aiming to meet the real-time detection requirements of wearable devices. Shen et al. [18] proposed a multi-scale dilated CNN incorporating an attention mechanism, using 60-s ECG segment as input and assigning the event category of that segment as the label.

A long-standing challenge in OSA detection is the severe imbalance between normal and respiratory events (apneas and hypopneas). Researchers have explored data augmentation and cost-sensitive learning to mitigate this issue [17], [19]. Attention mechanisms have also shown promise in enhancing model focus on salient signal segments. SE-MSCNN [20] network, which using 60-s ECG singal as input and the corresponding event category as the label. Mashrur et al. [21] input 60-s ECG scalograms generated via wavelet transform into a compact CNN, with the label likewise being the event category of that segment.

Meanwhile, deep learning has also demonstrated strong potential in automatic sleep staging research. This task primarily relies on EEG and commonly uses a single 30-s EEG epoch as input, with the corresponding sleep stage of that epoch as the label. DeepSleepNet [22] is a CNN-RNN framework that extracts features from raw single-channel EEG and uses bidirectional LSTM to capture temporal dynamics, achieving around 82% accuracy. SeqSleepNet [25] uses a sequence of 20 $\times$ 30-s EEG epochs (a total of 10 minutes) as input, where each input has its own corresponding 20 labels in total, and employs a hierarchical RNN to jointly model both local and global dependencies to improve the continuity.

Several subsequent studies have extended these frameworks. For example, DeepSleepNet-lite [26] offers a lightweight alternative to DeepSleepNet with competitive accuracy and reduced parameters. U-time [23] is a fully convolutional encoder-decoder network that produces dense predictions from multi-channel PSG signals and achieves good performence across multiple datasets. Overall, CNN-based local feature extractors and RNN-based sequence models have become standard in sleep staging, consistently improving benchmark performance.

B. Multi-Modal and Multi-Task Learning

Many studies have shown that combining multiple physiological signals enhances performance in both OSA detection and sleep staging. For instance, airflow, SpO2, and ECG have been identified as key signals for apnea and hypopnea classification [30], [31].

Traditional PSG analysis also uses EEG, EOG, and EMG jointly to distinguish sleep stages. XSleepNet [32] takes a 20 $\times$ 30-s (10 min) sequence as input, constructs EEG/EOG/EMG multi-perspective parallel branches and fuses them at the feature layer. U-time [23] also confirmed that multichannel input improves model performance to variations in montage and

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clinical recording conditions, which is critical for real-world deployment. Overall, multi-modal networks have consistently outperformed unimodal approaches by leveraging the synergy among different physiological signals.

Existing methods often ignore the physiological coupling between OSA and sleep stages, where OSA events trigger arousals and stage shifts, while sleep structure influences respiratory stability. A unified modeling framework could better capture these interactions.

EEG-CLNet [33] is a collaborative learning model based on single-channel EEG. To align the label granularity of the two tasks, they concatenated two consecutive 30-s EEG epochs into a 60-s sequence as the input; the sleep-staging label consists of the stage labels of the two 30-s epochs, while the OSA detection label is a single label for the corresponding 60 s window. The model jointly trains OSA detection and sleep staging via inter-task knowledge distillation and local parameter sharing. SleepFM [34] builds a base model using 19-channel signals and performs contrastive pre-training on data from over 14,000 participants. Full night recordings are segmented into consecutive 30-s epochs as the basic input; each epoch carries two labels, namely a sleep stage label and a binary OSA label, enabling representation transfer to downstream tasks such as sleep staging and OSA detection. Another study [35] designed a sequential model based on single-channel EEG, combining CNN and Transformer encoders. EEG is segmented into 30-s epochs and fed as a sequence to improve sleep staging, and intermediate features are further used to predict Apnea-Hypopnea Index (AHI). The study also conducted OSA detection at the 30-s epoch level, achieving a test accuracy of 67.68% , highlighting remaining challenges in epoch-level OSA detection. XSleepFusion [36] uses Flow, chest effort, and SpO2 as multi-modal inputs on the NSRR dataset; for each target 30-s segment, it appends 120-s of context on both sides to form a 270-s segment-level sample. The labels are the sleep stage and OSA event status of the central 30-s segment, and a dual-stage information-bottleneck mechanism is employed for interpretable modeling.

Although some studies have explored joint modeling of OSA detection and sleep staging, limitations remain: coarse multi-modal fusion, poor recognition of REM/N1, limited clinical validation, and the supervisory granularity of the two tasks is frequently misaligned. To address these issues, we propose MT-TASPPNet for systematic multi-modal, multi-task modeling. The model extracts multi-scale representations from neurophysiological and cardiorespiratory signals and employs a decoupled dual-task architecture to jointly perform sleep staging and OSA detection. Inspired by the label-alignment concept of EEG-CLNet [33], we retain the 30-s staging granularity while aligning supervision for both tasks to a common temporal anchor, and we incorporate longer temporal context to enhance cross-task synergy and temporal consistency.

III. Methods

Building on our previous work [37], this study extends that foundation to construct a multi-task model capable of simultaneously performing OSA detection and sleep staging. Compared to single-task learning (STL), MTL reduces parameter overhead and improves efficiency, while leveraging the physiological interplay between sleep stages and respiratory events. The proposed MT-TASPPNet retains the use of time–frequency representations of multi-modal inputs, incorporates both task-specific and shared ASPP-based modules, and leverages lightweight priors to enhance the recognition of subtle sleep states. The overall architecture of the proposed model is illustrated in Fig. 1.

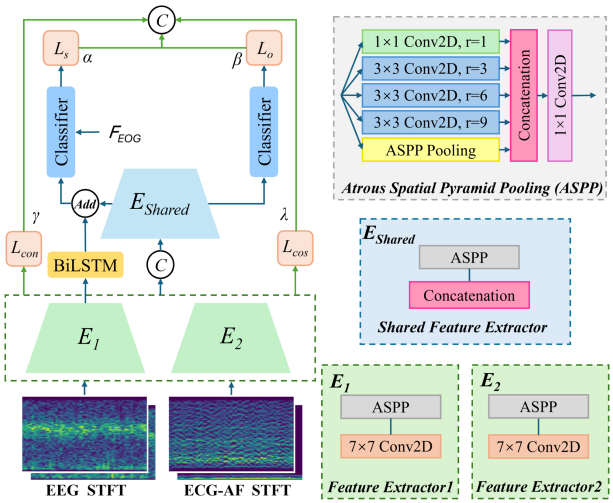


Fig. 1. Overview of the proposed MT-TASPPNet architecture.

A. Multi-Scale Feature Extraction

The backbone of MT-TASPPNet consists of two parallel feature extraction submodules, each dedicated to processing a pair of input signals: one for EEG (EEG-C3 and EEG-C4), and the other for ECG and airflow. This modular design preserves the TF structure of each modality while enabling efficient multi-scale contextual modeling. By modeling each signal pair independently, the network avoids early fusion interference and captures localized dynamics and multi-temporal patterns of different physiological activities more precisely.

Each single-channel input signal is represented as a tensor $x_t \in \mathbb{R}^{B\times T\times F}$, where $B = 128$ denotes the batch size, $T = 59$ is the number of time steps, and $F = 129$ is the number of frequency bins. For each pair (e.g., EEG-C3 and EEG-C4), the signals are concatenated along the channel axis to form a joint input of shape $X_i \in \mathbb{R}^{B\times 2\times T\times F}$, where $i \in \text{EEG}, \text{ECG-AF}$.

Each pair input is first passed through a 7×7 convolutional encoding block for low-level feature extraction and spatial downsampling. The convolutional block in this work follows the standard configuration:

$$\text{Conv2D} \rightarrow \text{BatchNorm2D} \rightarrow \text{ReLU} \tag{1}$$

The resulting feature maps are then processed by a two-dimensional ASPP module. The ASPP module comprises five parallel branches: one 1×1 atrous convolution with a dilation rate of 1, three 3×3 atrous convolutions with dilation rates of $3, 6, 12$, and one global average pooling branch. Each branch uses the same convolutional structure as above.

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This multi-branch architecture allows the ASPP module to capture both local and global features under different receptive fields without significantly increasing the number of parameters, making it well-suited for modeling the temporal dependencies and spectral variations in physiological signals.

Although the two submodules share an identical structure, their parameters are completely independent, enabling them to specialize for the statistical characteristics and spectral distributions of EEG and cardiopulmonary signals, respectively. After feature extraction, the two outputs are denoted as $\mathcal{F}_1, \mathcal{F}_2 \in \mathbb{R}^{B\times C'\times H\times W}$, where C' is the number of output channels, and H, W are the spatial dimensions after downsampling.

These feature maps are then concatenated along the channel axis to form a fused representation. The "extract-then-fuse" strategy enhances cross-modal representation capacity while maintaining the semantic diversity of each modality. The fused feature map is then forwarded to a shared ASPP module for further semantic abstraction, yielding the final shared representation: $\mathcal{F}_{\text{share}} = ASPP(Concat[\mathcal{F}_1, \mathcal{F}_2]) \in \mathbb{R}^{B\times C'\times H\times W}$. which is used for downstream multi-task modeling and classification.

B. EOG Feature Integration

To enhance the model's discriminative capacity in sleep staging—particularly in distinguishing between the N1 and REM stages—this study incorporates a prior statistical feature vector $f_{\text{EOG}} \in \mathbb{R}^{B\times de}$ extracted from the EOG-L signal. This vector is computed offline before training and participates in high-level semantic fusion during inference.

For each 1D input sample x_n with sampling frequency f_s and length N, we compute the following three slow-rolling eye movement-related features: the standard deviation reflecting global signal amplitude fluctuation, the frequency of eye movements per unit time, and the average duration between successive movements. These are combined to form the feature vector: $f_{\text{EOG}} = [\text{std}, \text{freq}, \text{dur}]$ The corresponding computations are defined as follows:

$$\begin{cases} \text{std} = \sqrt{\frac{1}{N}\sum_{n=1}^{N}(x_n - \bar{x})^2} \\ \text{freq} = \frac{Z}{T}, \quad T = \frac{N}{f_s} \\ \text{dur} = \frac{1}{M-1}\sum_{j=1}^{M-1}\frac{p_{j+1}-p_j}{f_s} \end{cases} \tag{2}$$

Here, $\bar{x}$ denotes the mean value of the signal; Z is the number of zero-crossings; M is the total number of detected peaks; and p_j represents the position (index) of the j-th peak. These features will be integrated with the backbone representation in Section III.C and used in the sleep staging branch to improve the model's performance in recognizing fine-grained sleep stages such as N1 and REM, while also enhancing physiological interpretability.

C. Multi-Task Prediction

To achieve joint modeling of OSA detection and sleep staging, two independent task-specific classification branches are constructed based on the shared feature representation $\mathcal{F}_{\text{share}}$. First, global average pooling is applied to compress the shared features into a unified low-dimensional representation.

For the sleep staging task, we further enhance temporal modeling by introducing a bidirectional long short-term memory (BiLSTM) network after the EEG branch feature $\mathcal{F}_1$. The BiLSTM takes the flattened temporal sequence as input, encodes contextual information in both forward and backward directions, and performs temporal average pooling to obtain a global sequence-level representation. This sequence feature is then fused with the pooled shared feature through residual addition, which enhances the temporal modeling capacity of the shared representation.

Finally, the BiLSTM-enhanced shared feature is concatenated with f_{EOG}, to form the fused representation: $\mathbf{z}_s = Concat[GAP(\mathcal{F}_{\text{share}}) + BiLSTM(\mathcal{F}_1), \quad f_{\text{EOG}}] \in \mathbb{R}^{B\times d_z}$, where $d_z = d_f + d_e$, with d_f being the dimension of $GAP(\mathcal{F}_{\text{share}}) + BiLSTM(\mathcal{F}_1)$.

The final sleep stage prediction is generated by a three-layer feedforward neural network $\phi(\cdot)$, expressed as:

$$\hat{\mathbf{y}}_s = \phi(\mathbf{z}_s; \{\mathbf{W}_s^{(l)}, \mathbf{b}_s^{(l)}\}_{l=1}^{3}) \in \mathbb{R}^{B\times N_e\times C_s} \tag{3}$$

where N_e denotes the number of sequential segments per sample, and C_s is the number of categories of sleep stages.

For the OSA detection task, no EOG features are used. Instead, the pooled shared feature is directly modeled, and the classification output is produced by another three-layer feedforward network ϕ_o, as follows:

$$\hat{\mathbf{y}}_o = \phi_o(GAP(\mathcal{F}_{\text{share}}); \{\mathbf{W}_o^{(l)}, \mathbf{b}_o^{(l)}\}_{l=1}^{3}) \in \mathbb{R}^{B\times C_o} \tag{4}$$

where C_o enotes the number of output classes for OSA detection.

This multi-task design enables shared feature utilization while maintaining structural decoupling between tasks. It provides each task with an independent high-level semantic space, enhancing overall model performance and facilitating collaborative optimization across tasks.

D. Joint Loss

To achieve joint optimization of the two tasks, namely OSA detection and sleep staging, within the MTL framework, we design a composite loss function $\mathcal{L}_{\text{total}}$that balances discriminative representation, inter-modality consistency, and task performance. The two modality-specific feature extractors produce time-frequency (TF) feature maps $\mathcal{F}_1$ and $\mathcal{F}_2$, upon which the following three loss components are constructed:

First, to enhance the discriminability of different modalities in the high-level semantic space, we introduce a contrastive learning loss $\mathcal{L}_{\text{con}}$. Let $\mathbf{f}_1 = \text{GAP}(\mathcal{F}_1)$ and $\mathbf{f}_2 = \text{GAP}(\mathcal{F}_2)$ denote the pooled feature representations of the two modalities. A bidirectional InfoNCE loss is applied as follows:

$$\mathcal{L}_{\text{con}} = \frac{1}{2}\cdot\left(\ell_{\text{ce}}\left(\frac{\mathbf{f}_1\cdot\mathbf{f}_2^{\top}}{\tau}, \mathbf{I}\right) + \ell_{\text{ce}}\left(\frac{\mathbf{f}_2\cdot\mathbf{f}_1^{\top}}{\tau}, \mathbf{I}\right)\right) \tag{5}$$

Here, τ is a temperature parameter, $\mathbf{I}$ is the identity label matrix used to construct pseudo-labels between matching samples, and ℓ_{ce} denotes the cross-entropy loss function.

Next, to encourage directional consistency between the proxy representations of the two modalities in the output space, we introduce the cosine similarity loss $\mathcal{L}_{\text{cos}}$. The proxy vectors are

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obtained via linear transformation of the pooled features as:

$$\mathbf{p}_i = \mathbf{f}_i \mathbf{W} i + \mathbf{b} i, \quad i \in 1, 2 \tag{6}$$

The overall cosine loss is then computed as:

$$\mathcal{L}_{\cos} = \frac{1}{B} \sum_{i=1}^{B} \left(1 - \cos \left(\frac{\mathbf{p}_1^{(i)}}{\|\mathbf{p}_1^{(i)}\|_2}, \frac{\mathbf{p}_2^{(i)}}{\|\mathbf{p}_2^{(i)}\|_2} \right) \right) \tag{7}$$

This term minimizes the average cosine distance between the sample-level proxy vectors, guiding the feature exctractors to maintain consistent semantic directions.

Third, we introduce the task classification loss $\mathcal{L}_{\text{ce}}$, which comprises weighted cross-entropy losses for the sleep staging and OSA detection tasks. Let $\hat{\mathbf{y}}_s, \hat{\mathbf{y}}_o$ denote the predictions, $\mathbf{y}_s, \mathbf{y}_o$ the corresponding ground truths, and $\mathbf{w}s, \mathbf{w}o$ the class weights used to address class imbalance. The loss is defined as:Here, the hyperparameters α and β control the relative weights of the two task losses to ensure balanced optimization.

Finally, the joint optimization objective $\mathcal{L}_{\text{total}}$ combines the above three components as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{ce}} + \gamma \cdot \mathcal{L}_{\cos} + \lambda \cdot \mathcal{L}_{\text{con}} \tag{9}$$

where γ and λ are weighting coefficients for the cosine consistency loss and contrastive loss, respectively. This joint optimization strategy enables both cross-modality coupling and decoupling, balancing discrimination and alignment, and effectively enhances the stability and adaptability of the multi-task framework.

IV. Experiment

A. Datasets

To ensure the stability and adaptability of the proposed model, we utilized large-scale sleep datasets from both public repositories (via NSRR [39]) and clinical sources. Each dataset includes subjects with varying degrees of OSA severity as well as healthy individuals.

1) SHHS Dataset: The Sleep Heart Health Study (SHHS) is a multi-center cohort study investigating the cardiovascular and respiratory implications of sleep disorders [40]. It consists of two subsets: SHHS1, which includes 6,441 recordings collected between 1995 and 1998, and SHHS2, comprising 3,295 recordings collected between 2001 and 2003. These datasets contain comprehensive physiological signals including EEG (C3/A2, C4/A1), EOG, EMG, ECG, airflow, thoracic and abdominal effort, and more. Gender and age information were available for the majority of subjects, with SHHS1 including 2,744 males and 3,019 females, and SHHS2 including 1,226 males and 1,425 females. In SHHS1, EEG and ECG were sampled at 125 Hz, EOG at 50 Hz, and Airflow at 10 Hz; in SHHS2, the sampling rates were identical to those in SHHS1, except that ECG was recorded at 250 Hz

2) Sydney Sleep Biobank: The Sydney Sleep Biobank (SSB) is Australia's first certified sleep biobank [41]. It comprises data collected across three clinical sites and includes PSG recordings with accompanying scoring information, biospecimens (e.g., DNA, proteins, and metabolites), questionnaires on symptoms, lifestyle factors, and comorbidities, as well as partial craniofacial imaging. All participants were undiagnosed with OSA at the time of enrollment. In this study, we specifically utilized the subset of PSG data collected from Royal North Shore Hospital between 2020 and 2022, referred to as **RNS**. Out of 830 initial recordings, 778 high-quality PSGs were retained after excluding those with missing channels or insufficient duration. The retained recordings include EEG (C3/M2, C4/M1), EOG, EMG, ECG, airflow, and body position signals, enabling both sleep staging and OSA detection tasks. Among the participants, 648 had complete gender data (388 male, 260 female), while 129 records lacked age, BMI, or gender information. EEG, EOG, and ECG were originally sampled at 256 Hz, whereas airflow was sampled at 32 Hz

All data collection procedures were approved by the Northern Sydney Local Health District Human Research Ethics Committee (HREC) under protocol number HREC/17/HAWKE/340, with site-specific approval for Royal North Shore Hospital (SSA/18/HAWKE/127).

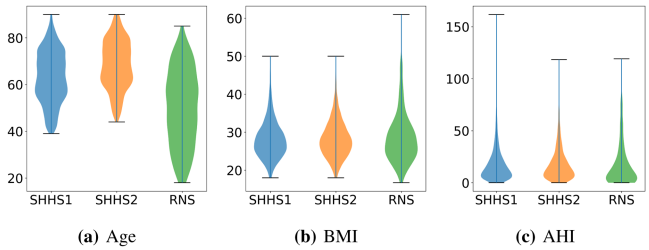


Fig. 2. Comparison of (a) Age, (b) BMI, and (c) AHI across three datasets. Vertical black lines denote the minimum and maximum values.

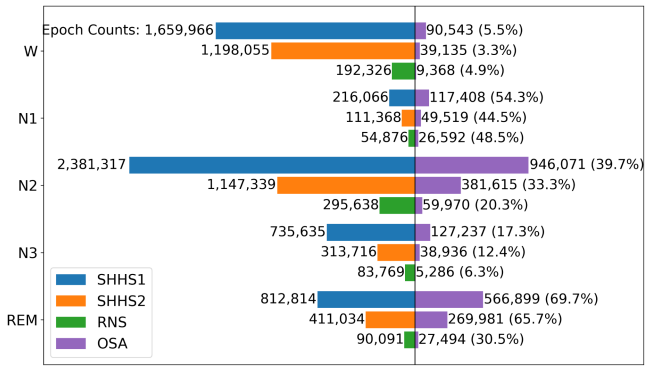


Fig. 3. Distribution of total and OSA-specific epoch counts across sleep stages in the SHHS1, SHHS2, and RNS datasets.

Fig. 2 presents the distributions of Age, BMI, and AHI across the three datasets. They are derived from clinical records: Age and BMI from intake examinations, AHI from AASM-based clinical scoring; for SHHS, the AHI variable is ahi_a0h3a.

Fig. 3 presents the sleep stage composition and the proportion of OSA-related epochs in SHHS1, SHHS2, and RNS. OSA events are more frequent in N1, N2, and REM stages, while relatively rare during wake and N3 periods.

B. Pre-Processing

This study addresses both OSA detection and sleep staging, requiring modality-specific preprocessing to construct a unified

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TABLE I
SUBJECT AND EPOCH-LEVEL DISTRIBUTION OF DATA PARTITIONS ACROSS DATASETS

Dataset	Split	Group (Subjects)				OSA Epochs		Sleep Staging Epochs				
	(7:2:1)	Non-OSA	Mild	Moderate	Severe	Normal	OSA	Wake	N1	N2	N3	REM
SHHS1	Train(4034)	62	677	1700	1595	1388561	645294	1163657	153637	1662324	517821	570271
	Valid(1153)	11	212	460	470	392921	186323	326278	41699	477334	163800	149377
	Test(576)	8	81	254	233	197338	92462	170031	20730	241659	68437	78743
SHHS2	Train(1855)	47	469	774	565	839501	271704	834289	77468	803287	286957	220409
	Valid(531)	22	125	209	175	240598	78543	248078	226928	21788	60935	80553
	Test(265)	3	74	117	71	121064	39346	115688	12112	117124	32372	43524
RNS	Train(544)	193	145	102	104	206400	44344	13552	37179	207401	58482	62874
	Valid(156)	56	41	29	30	58770	13274	36626	11884	60036	18541	17001
	Test(78)	27	21	15	15	28825	6737	20148	5813	28201	8286	8676

multi-task input. For OSA detection, ECG and airflow signals are denoised using low-pass and band-pass filters to remove baseline drift and isolate respiratory rhythms. For sleep staging, EEG and EOG channels undergo band-pass and notch filtering to eliminate noise and preserve relevant neural activity. All signals are standardized using Z-score normalization on a per-subject basis.

The labeling follows the EEG-CLNet [33] paradigm and adheres to the AASM [5] event definition (apnea/hypopnea duration $\geq$ 10s) and the AHI calculation convention (see Eq. (10)). In implementation, we use a 1-min target segment and map timestamped events to a binary segment-level label, thereby aligning supervision across tasks and maintaining consistency with the AHI and Total Sleep Time (TST) definition. Sleep staging retains the clinical 30-s granularity: for each 1-min target segment, we concatenate the two constituent 30-s epochs and supervise each with its corresponding stage label.

To unify the supervision granularity of the two tasks and enhance cross-task synergy, the input comprises a 1-min target segment with $\pm$ 1-min context (total length 3 min) and a 1-min stride. This setting preserves the 30-s temporal resolution of staging labels while introducing longer temporal context, reducing optimization conflicts and improving the stability and discriminative power of joint learning. All signals are converted into log-scaled magnitude spectrograms using STFT with Hamming windowing to reduce spectral leakage. Each input X_i maps to multi-task labels: $X_i \rightarrow \{y_o^i, \mathbf{y}s^i\}$, where $y_o^i \in O, N$, and $\mathbf{y}_s^i = [y_{s,1}^i, y_{s,2}^i]$, $y_{s,j}^i \in \{W, N1, N2, N3, REM\}$ denotes the two merged sleep stages.

C. Evaluation Criteria

We adopted standard evaluation metrics from the STL setting to assess the performance of the two tasks independently. All metrics were computed at the epoch level.

For the OSA detection task, we report Accuracy (Acc), Sensitivity (Sen), Specificity (Spe), macro-averaged F1 (MF1), class-wise precision for the Normal and OSA [Prec.0, Prec.1], and area under the curve (AUC).

For sleep staging, we followed previous studies [33], [42] and employed three overall metrics: Acc, MF1, Cohen's kappa ($\mathcal{K}$), and AUC, together with per-stage F1 scores for W, N1, N2, N3, and REM to enable detailed class-wise analysis.

D. Implementation Details

The model was trained for 60 epochs using the Adam optimizer with a fixed learning rate of $1e-4$ and a batch size of 128. The checkpoint achieving the best validation MF1 was used for final evaluation. The overall training objective combined OSA detection and sleep staging losses with weights of 3.0 and 0.7, respectively, together with a 0.2-weighted cosine loss and a 0.1-weighted contrastive loss on the shared representations. The subject- and epoch-level splits for training, validation, and testing across all datasets are summarized in Table I. Dropout was applied to mitigate overfitting. All experiments were implemented in PyTorch and executed on an NVIDIA RTX 4090 GPU with mixed-precision training.

To quantify model complexity, THOP was used to compute the parameter count and computational cost. The model has approximately 22.75 million parameters and 10.17 GFLOPs. On an RTX 4090 GPU, the average inference time for a 1-minute sample is 1.23 ms, corresponding to a throughput of 811.26 samples/s, demonstrating a favorable balance between accuracy and efficiency.

V. RESULTS

To comprehensively evaluate the performance of the proposed MT-TASPPNet, we present its multi-task classification results on three large-scale datasets and further conduct ablation studies to analyze the contribution of each key component. We focus on metrics such as Acc, MF1 and AUC for both tasks to evaluate the model's stability and adaptability under different datasets and task settings.

A. Multi-Task Classification Performance

1) Overall Results of MT-TASPPNet: Table II presents the overall performance of MT-TASPPNet. Overall, the model demonstrates good generalization and task balance across different datasets and tasks, with particularly strong multi-modal and multi-task feature modeling capabilities observed on the SHHS2 and RNS datasets.

For the OSA detection task, the model achieves accuracy scores of 0.868 and 0.884, and F1 scores of 0.821 and 0.814 on SHHS2 and RNS respectively—both outperforming the results

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TABLE II
MULTI-TASK PERFORMANCE OF DIFFERENT DATABASES ON MT-TASPPNET

Dataset	OSA Detection					Sleep Staging								
	Acc	MF1	Prec.0	Prec.1	AUC	Acc	MF1	F1-W	F1-N1	F1-N2	F1-N3	F1-REM	$\mathcal{K}$	AUC
SHHS1	0.798	0.772	0.837	0.673	0.777	0.828	0.744	0.91	0.428	0.842	0.712	0.829	0.756	0.845
SHHS2	0.868	0.821	0.911	0.733	0.82	0.834	0.749	0.935	0.436	0.825	0.775	0.772	0.77	0.847
RNS	0.884	0.814	0.923	0.686	0.82	0.776	0.735	0.858	0.466	0.789	0.744	0.816	0.697	0.847

on SHHS1 (accuracy: 0.798, F1: 0.772). This indicates stronger discriminative ability on higher-quality datasets.

For the sleep staging task, MT-TASPPNet achieves the highest overall accuracy (0.834) and AUC (0.847) on SHHS2. Although the SHHS2 test set contains nearly equal numbers of REM and N3 epochs, their F1 scores are also very close, which contrasts with the performance trends observed in SHHS1 and RNS. We speculate that this may be related to the imbalance in the SHHS2 training set, where N3 epochs significantly outnumber REM epochs, potentially limiting the model's ability to generalize REM-specific representations.

On the RNS dataset, although the F1-W is relatively lower (0.858), the F1-N1 stage reaches 0.466, higher than in the other two datasets, demonstrating the model's effectiveness in recognizing marginal stages in real-world clinical scenarios. It is worth noting that many wake periods in RNS are labeled during signal dropout episodes (e.g., when participants leave the bed during the night), which may contribute to the lower wake-stage accuracy. This lower performance does not reflect the model's limitations, but rather the annotation inconsistencies in the dataset.

To assess cross-dataset generalization, we used the model trained on SHHS1 to evaluate performance on the full SHHS2 and RNS datasets. In the SHHS1 $\rightarrow$ SHHS2 setting, the sleep staging accuracy reached 0.832, with a MF1 of 0.728 and a $\mathcal{K}$ of 0.760. In the SHHS1 $\rightarrow$ RNS setting, performance further declined to 0.772 accuracy, 0.698 MF1, and $\mathcal{K}$ of 0.688. These results highlight the challenges of cohort generalization under varying population characteristics and recording conditions.

2) Individual-Level AHI Prediction Performance: As part of the overall performance evaluation of MT-TASPPNet, we further analyzed the model's ability to estimate AHI at the individual level. Following the clinical definition of it, we compute each subject's AHI based on the number of predicted OSA segments and TST, which excludes epochs labeled as wake. Specifically, AHI is computed as:

$$AHI = \frac{60 \times OSA\ epoches}{TST} \tag{10}$$

This approach reflects clinical practice and allows indirect assessment of the model's effectiveness in capturing the frequency of apnea-hypopnea events. Fig. 4 presents the AHI prediction performance across three datasets. Fig. 4(b) compares the minute-level AHI derived from our labeling rules (Ground Truth) with the clinical AHI, while Fig. 4(b) shows the comparison between the final predicted AHI and the Ground Truth, including both scatter and Bland–Altman plots. The mean absolute error (MAE) for each dataset is also reported to quantify individual-level prediction accuracy.

Given the relatively large prediction error in SHHS1, we further examined its event label files and documentation. We found that Hypopnea events in SHHS were annotated according to earlier standards, differing from the widely adopted AASM 2012 or 2017 guidelines [5]. Many of these events lack associated oxygen desaturation or arousal, and therefore require filtering before being used for accurate clinical AHI calculation (10). Since no such filtering was applied in our study, events inconsistent with modern standards may have contributed to the increased prediction error. For example, the subject , "shhs1-202250" has a clinical AHI (ahi_a0h3a) of 1.96, while the corresponding label file includes 93 Hypopnea events, far more than expected for that AHI level.

In contrast, our RNS dataset achieved the lowest MAE in the comparison between minute-level AHI and clinical AHI (Fig. 4(a)), further supporting the reliability of our labeling rule and the consistency of RNS annotations with current clinical standards.

As shown, the predicted AHI values exhibit strong agreement with the ground truth in the SHHS2 and RNS datasets, achieving Pearson correlation coefficients of $r = 0.84$ and $r = 0.89$, respectively, indicating good performance. In contrast, the SHHS1 dataset shows slightly lower correlation $r = 0.65$, but still maintains acceptable consistency.

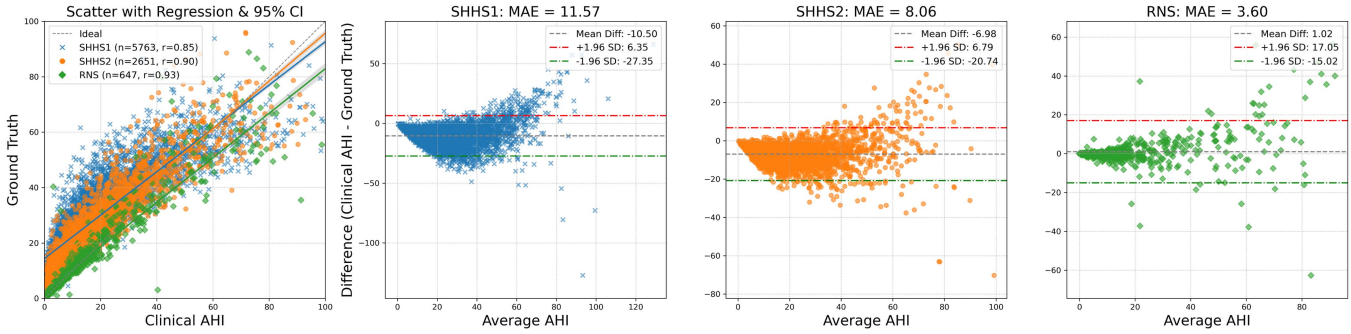
The Bland–Altman plots further illustrate the bias and agreement between the predicted and ground truth AHI. SHHS2 and RNS show smaller mean differences and narrower limits of agreement ($\pm$ 1.96 SD), suggesting stable and reliable model predictions.

To intuitively assess the model's ability to identify individual OSA severity, subjects were grouped by clinical AHI thresholds into Non-OSA (<5), Mild (5–15), Moderate (15–30), and Severe (≥ 30). Confusion matrices comparing predicted and true severity levels are shown in Fig. 5. The model exhibited relatively stable performance in identifying moderate and severe OSA cases, achieving correct classification rates above 60% in the SHHS2 and RNS datasets.

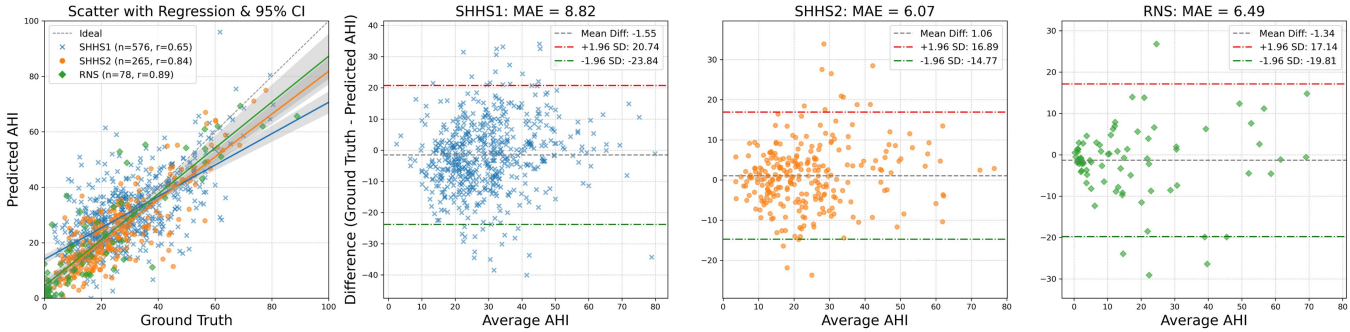
3) Ablation Experiment of MT-TASPPNet: We conducted two groups of ablation experiments on the RNS clinical dataset to assess the contribution of each component and the effect of the input window design on the multi-task performance. We chose RNS because its clinical setting and annotation quality are high and its moderate size allows stable replication and clear trend observation.

The first is module-level ablation (see Table III). The gradual removal of specific modules resulted in varying degrees of

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(a) Minute-level AHI derived from our labeling rules (Ground Truth) versus clinical AHI.



(b) Final predicted AHI versus Ground Truth AHI.

Fig. 4. Visual comparison of AHI agreement across SHHS1, SHHS2, and RNS datasets. For each dataset, scatter plots with regression lines and 95% confidence intervals are shown alongside Bland–Altman plots. Here, n indicates the number of subjects, and r denotes the Pearson correlation coefficient. The black dashed line represents the mean difference (Mean Diff), and the green and red dashed lines indicate the $\pm$ 1.96 standard deviation (SD) limits.

TABLE III
ABLATION OF MT-TASPPNET COMPONENTS ON THE RNS DATASET

Module			$\mathcal{L}$			OSA Detection					Sleep Staging								
EOG	LSTM	Share Extractor	$\mathcal{L}_{cos}$	$\mathcal{L}_{con}$	$\mathcal{L}_{ce}$	Acc	MF1	Prec.0	Prec.1	AUC	Acc	MF1	F1-W	F1-N1	F1-N2	F1-N3	F1-REM	$\mathcal{K}$	AUC
	✓	✓	✓	✓	✓	0.865	0.786	0.91	0.637	0.793	0.761	0.712	0.839	0.459	0.782	0.721	0.758	0.675	0.827
✓		✓	✓	✓	✓	0.842	0.77	0.935	0.563	0.8043	0.753	0.706	0.84	0.456	0.782	0.68	0.77	0.663	0.819
✓	✓		✓	✓	✓	0.863	0.777	0.914	0.643	0.775	0.775	0.711	0.871	0.424	0.802	0.692	0.768	0.692	0.816
		✓	✓	✓	✓	0.892	0.81	0.919	0.751	0.797	0.729	0.685	0.832	0.432	0.75	0.656	0.755	0.633	0.815
✓	✓	✓		✓	✓	0.872	0.796	0.927	0.652	0.8	0.728	0.68	0.838	0.407	0.731	0.692	0.742	0.635	0.819
✓	✓	✓	✓		✓	0.873	0.803	0.935	0.645	0.818	0.781	0.731	0.86	0.441	0.797	0.745	0.81	0.701	0.834
✓	✓	✓			✓	0.877	0.797	0.92	0.681	0.793	0.766	0.716	0.87	0.466	0.786	0.683	0.776	0.681	0.825
✓	✓	✓	✓	✓	✓	0.884	**0.814**	0.923	**0.686**	**0.82**	0.776	**0.735**	0.858	**0.466**	0.789	0.744	**0.816**	0.697	**0.847**

performance degradation. Removing the EOG modality leads to a clear drop in N1 recognition (F1–N1 decreases from 0.466 to 0.459) and a decline in overall OSA detection performance (e.g., MF1), indicating that eye-movement information provides complementary cues for light sleep and OSA events. In contrast, adding EOG noticeably improves F1–N1, confirming its value for boundary-stage recognition. After removing the LSTM, F1–N1 further falls to 0.424, demonstrating its effectiveness in modeling temporal dependencies and stage transitions. The shared feature extractor is also critical: eliminating it reduces Prec.1 (OSA) and F1–N2, showing that a shared semantic space facilitates cross-task knowledge transfer. Additionally, loss function configurations were validated. Removing either $\mathcal{L}_{cos}$ or $\mathcal{L}_{con}$, and retaining only $\mathcal{L}_{ce}$, led to performance drops, especially in Prec.1 and F1-N1. Overall, the full configuration achieves the best results (best values are bolded in the table).

We then performed ablation on the input segment length in Table IV. Keeping the architecture, loss terms, and training schedule fixed, we vary only the input segment length using four settings: 30-s, 1 min, 3 min, and 5 min. The stride is fixed at 1 min, and the label-alignment policy remains unchanged (OSA uses a binary label for the target 1-min segment; sleep staging uses the two 30-s stage labels within that target minute). Specifically, 30-s uses no context; 1 min uses the target minute only; 3 min and 5 min append $\pm$ 1 min and $\pm$ 2 min of context, respectively. The results show that 3 min achieves the best or near-best performance on both tasks, while 5 min is slightly inferior to 3 min; 30-s yields marginally higher staging

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TABLE IV
ABLATION ON INPUT SEGMENT LENGTH ON THE RNS DATASET

Input Segment	OSA Detection					Sleep Staging								
Length	Acc	MF1	Prec.0	Prec.1	AUC	Acc	MF1	F1-W	F1-N1	F1-N2	F1-N3	F1-REM	$\mathcal{K}$	AUC
30-s	0.842	0.731	0.888	0.608	0.718	0.788	0.726	0.894	0.435	0.815	0.722	0.765	0.708	0.821
1 min	0.875	0.784	0.909	0.701	0.768	0.762	0.713	0.872	0.426	0.783	0.722	0.762	0.68	0.833
3 min	**0.884**	**0.814**	**0.923**	0.686	**0.82**	0.776	**0.735**	0.858	**0.466**	0.789	**0.744**	**0.816**	0.697	**0.847**
5 min	0.876	0.77	0.894	0.754	0.739	0.736	0.672	0.835	0.41	0.782	0.606	0.725	0.636	0.787

TABLE V
PERFORMANCE COMPARISON OF DIFFERENT MODELS ON RNS DATASET

Model	Year	Mode	OSA Detection					Sleep Staging								
			Acc	MF1	Prec.0	Prec.1	AUC	Acc	MF1	F1-W	F1-N1	F1-N2	F1-N3	F1-REM	$\mathcal{K}$	AUC
DeepSleepNet [22]	2017	STL	0.858	0.790	**0.941**	0.597	**0.823**	0.793	**0.740**	0.876	0.461	**0.821**	0.730	0.811	0.715	0.832
U-time [23]	2019	STL	0.879	0.787	0.906	0.730	0.765	0.790	0.729	0.874	0.427	0.818	0.715	0.800	0.700	0.824
IITNet [24]†	2020	STL	0.809	0.449	0.810	0.188	0.500	0.615	0.510	0.786	0.238	0.636	0.635	0.259	0.467	0.695
DeepSleepLite [26]	2021	STL	0.867	0.798	0.938	0.625	0.821	0.774	**0.740**	0.845	0.422	0.820	0.735	0.758	0.690	0.82
SEMSCNN [20]	2022	STL	0.847	0.751	0.905	0.596	0.751	0.771	0.695	0.858	0.289	0.799	0.690	0.839	0.680	0.799
L-SeqSleepNet [27]†	2023	STL	0.857	0.773	0.919	0.615	0.781	0.771	0.712	0.860	0.381	0.801	0.673	0.844	0.684	0.822
BAFNet [28]†	2023	STL	0.724	0.628	0.883	0.357	0.664	0.755	0.672	0.840	0.250	0.793	0.703	0.773	0.657	0.788
SleepDG [29]	2024	STL	0.875	0.806	0.934	0.656	0.819	0.762	0.701	0.845	0.356	0.796	0.665	**0.841**	0.669	0.804
MT-TASPPNet	2025	STL	0.873	0.760	0.889	**0.756**	0.726	**0.794**	**0.740**	**0.880**	0.456	0.818	**0.748**	0.797	**0.718**	0.838
CNN-LSTM [38]	2025	MTL	0.810	0.447	0.810	0	0.500	0.636	0.740	0.694	0.187	0.730	0	0.672	0.474	0.696
CNN-LSTM [38]‡	2025	MTL	0.823	0.733	0.911	0.528	0.753	0.655	0.740	0.710	0.308	0.731	0.673	0.444	0.521	0.753
MT-TASPPNet	2025	MTL	**0.884**	**0.814**	0.923	0.686	0.820	0.776	0.735	0.858	**0.466**	0.789	0.744	0.816	0.697	**0.847**

† No mixed-precision training used.
‡ 5× weighted sampling of OSA samples.

consistency ($\mathcal{K} = 0.708$) but is overall worse than 3min on OSA metrics. Considering accuracy and computational cost, we adopt a 3min input window with a 1min stride as the default setting for the whole signal processing.

B. Compared With State-of-the-Art Methods

To validate the effectiveness of the proposed MT-TASPPNet, we conducted a comprehensive evaluation on the clinical RNS dataset, assessing its performance on both OSA detection and sleep staging, and comparing it against a variety of state-of-the-art deep learning methods. Compared to publicly available datasets such as SHHS, RNS contains fewer samples, but it more closely resembles real-world clinical deployment scenarios, making it a more suitable benchmark for evaluating model generalizability and practical applicability.

Table V summarizes the performance of all models across both tasks under STL and MTL paradigms. The best value for each metric is highlighted in bold, and the results of our proposed MT-TASPPNet are denoted with an underline.

To ensure a fair comparison, all baseline models were trained and evaluated using the same input modalities and a unified preprocessing pipeline as described in this work. For 1D network architectures (e.g., DeepSleepNet, SE-MSCNN, BAFNet), we performed upsampling to ensure that each input segment had a uniform length of 6000 samples. Additionally, as most baseline methods were originally designed for sleep staging with a five-class output structure, we modified their output layers to support binary classification for the OSA detection task.

All models were evaluated using their public implementations and were trained, validated, and tested under the same data split and training configurations used in this paper, minimizing the impact of external factors on performance comparison.

DeepSleepNet [22] and DeepSleepLite [26]: Typical CNN-LSTM frameworks originally developed for sleep staging. We modified their classifiers and independently evaluated their performance on both tasks.

U-time [23]: A fully convolutional U-Net-based architecture designed for sleep staging. We adapted it to accept 1D inputs and trained it separately for the two tasks.

IITNet [24]: This model utilizes a multi-stage framework combining ResNet and LSTM. We retained the single input head structure and concatenated our input signals along the channel dimension.

SE-MSCNN [20]: Initially proposed for OSA detection based on R-R intervals (RRI) and R-peak amplitudes (RPA). To ensure input consistency across models, we replaced its original input with the two signal modalities used in this study.

L-SeqSleepNet [27]: We adopted the optimal configuration reported in the paper (sequence length = 200, B = 20, K = 10) to capture temporal dependencies, and trained it under the STL paradigm.

BAFNet [28]: A lightweight 1D convolutional network using RRI and RPA as input. We replaced its input with the two signal modalities used in this work, enabling evaluation on both OSA detection and sleep staging tasks.

SleepDG [29]: A domain generalization STL method originally trained on SHHS data. In this study, we re-trained and evaluated the model exclusively on the RNS dataset.

CNN-LSTM [38]: A recently proposed MTL framework that integrates CNN-LSTM based feature extraction for joint prediction of sleep stages and respiratory events. We evaluated

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both its original implementation and a class-balanced version with 5× oversampling of OSA-positive samples.

MT-TASPPNet (STL): Our proposed architecture trained separately on OSA detection and sleep staging tasks in the single-task setting.

MT-TASPPNet (MTL): The full multi-task version of our proposed model, featuring modality-specific encoders, a multi-scale ASPP module, and decoupled task-specific classifiers.

1) Performance Analysis of Single-Task Model: Under the single-task setting, most baseline models exhibited promising performance in both OSA detection and sleep staging. For instance, SleepDG achieved the highest MF1 (0.806) in OSA detection. DeepSleepNet also showed outstanding performance in major sleep stages, with F1-W and F1-N2 reaching 0.876 and 0.821, respectively, indicating its solid capabilities in temporal modeling and sleep structure recognition. DeepSleepNetLite delivered stable performance across both tasks, highlighting its practical potential.

However, these conventional single-task models still have certain limitations, especially when dealing with transitional stages and imbalanced classes. For example, SE-MSCNN, a model specifically tailored for OSA detection, achieved a relatively low Prec.1 of 0.596, while L-SeqSleepNet, despite its acceptable performance in dominant stages like REM, yielded a relatively low F1-N1 score of 0.381.

In contrast, our proposed MT-TASPPNet demonstrated strong modeling capabilities even in single-task mode. Specifically, it achieved an MF1 of 0.760 in OSA detection and an F1-N1 of 0.456 in sleep staging, outperforming most single-task baselines. These results validate its reliable performance and effectiveness in capturing complex sleep structures and abnormal events. Notably, the model architecture is designed to accommodate both multi-modal signal integration and deep semantic representation, laying a solid foundation for subsequent multi-task learning.

2) Performance Analysis of Multi-Task Model: Under the multi-task setting, MT-TASPPNet further enhanced overall performance through shared low-level feature extraction and task-specific decoupled branches. The model achieved the best performance across multiple key metrics, including MF1 (0.814), F1-N1 (0.466), F1-REM (0.816), Kappa (0.697), and AUC (0.847), reflecting improved outcomes for both tasks and reduced performance gaps between them. This highlights the model's strong capability for collaborative task learning.

Importantly, the model did not rely on any data augmentation, resampling, or post-processing strategies during training, yet still maintained robust performance under highly imbalanced conditions, such as the scarcity of OSA-positive samples. These results stem from our proposed modality-aware design, multi-scale atrous convolution modules, and task-decoupling strategies, which jointly enable the model to retain critical information, mitigate task interference, and enhance recognition of transitional stages and rare events.

To better understand the trade-off between tasks in our network, we investigated the effect of different loss weight settings for two tasks. Specifically, we evaluated four OSA/Stage weight combinations: (1.0/1.0), (2.0/0.7), (3.0/0.7), and (3.0/1.0). Among them, the setting of (3.0/0.7) yielded the best overall performance, achieving the highest MF1 and AUC for OSA detection, while maintaining competitive sleep staging performance. These results suggest that placing a slightly higher emphasis on OSA detection can enhance this clinically critical task without substantially compromising staging accuracy. The minor drop in staging performance indicates an acceptable level of negative transfer, likely due to the higher weight assigned to the more imbalanced OSA detection.

VI. Discussion

MT-TASPPNet demonstrates high-accuracy multi-task sleep analysis while balancing architectural generalizability and cross-task synergy.

A. Distribution Analysis of Individual-Level Accuracy

To further assess the individual-level performance stability of MT-TASPPNet, we plotted the boxplots shown in Fig. 6, which illustrate the distribution of Accuracy and MF1 scores for OSA detection and sleep staging across the SHHS1, SHHS2, and RNS datasets. Overall, for OSA detection, most individuals achieved accuracy between 80% and 95% , and MF1 scores generally ranged from 75% to 90% , reflecting strong discrimination capability and stability. The RNS dataset exhibited higher individual accuracy, likely because the test set included some healthy subjects. The model successfully identified these non-OSA cases, leading to elevated accuracy at the subject level.

In contrast, performance in sleep staging showed slightly lower accuracy distributions (typically between 75% and 88%) and broader MF1 variability (approximately 60% to 85%), with RNS showing particularly large variation. This suggests that sleep stage classification is more sensitive to inter-subject variability. These results confirm that MT-TASPPNet maintains strong individual-level performance and demonstrates robust multi-task generalizability across diverse populations.

B. Individual-Level Visualization of Task Collaboration

To further evaluate the interpretability and temporal alignment of our model in practical clinical contexts, we provide individual-level visualizations of multi-task predictions. Fig. 7 presents case-level visualizations for four representative subjects, corresponding to healthy, mild, moderate, and severe OSA conditions. Each subplot simultaneously displays the predicted and true AHI values, along with a timeline-based alignment of five sleep stages (W, N1, N2, N3, REM) and the distribution of corresponding OSA events. This layout facilitates intuitive assessment of the temporal consistency and potential interactions between the two tasks.

As shown in Fig. 7(a), the model successfully identifies the normal sleep architecture of a healthy subject and correctly predicts a minimal or zero AHI, reflecting a low false positive tendency. Figs. 7(b) and 7(c) correspond to mild and moderate OSA cases, respectively. The predicted AHI values closely match the ground truths, and the model effectively detects clustered OSA events in light sleep stages, while maintaining low

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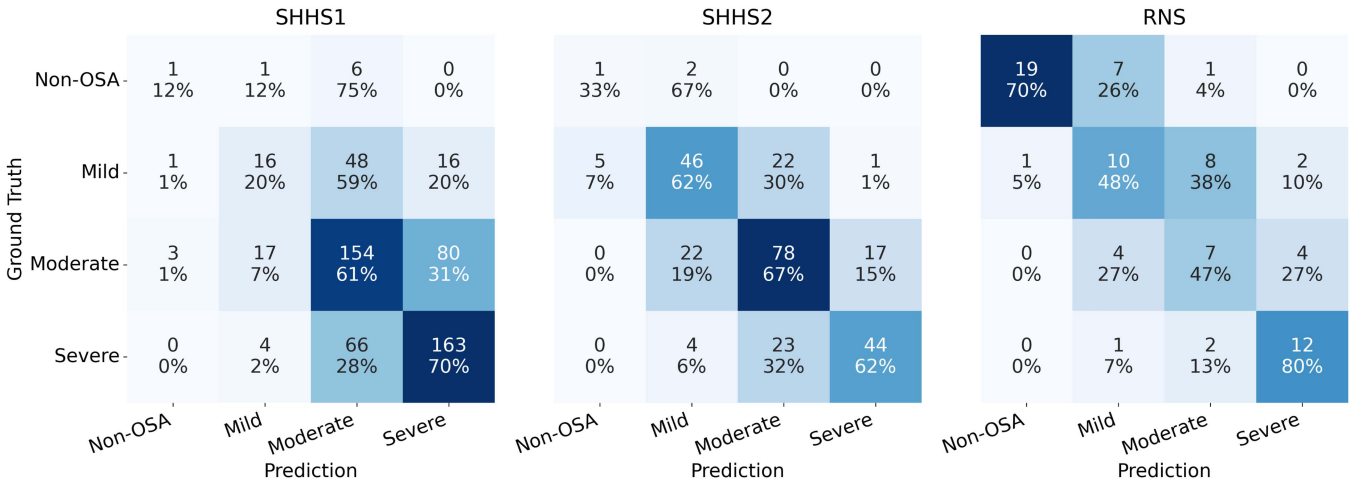
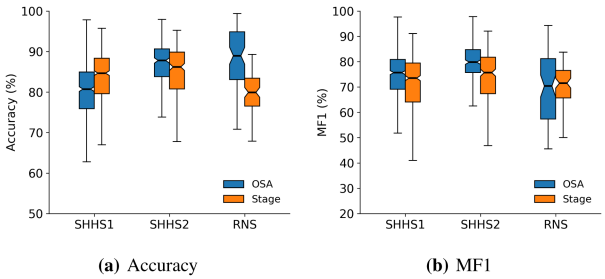


Fig. 5. Confusion matrices of predicted and true OSA severity levels for SHHS1, SHHS2, and RNS datasets.



(a) Accuracy (b) MF1

Fig. 6. Individual-level performance distribution of MT-TASPPNet on three datasets. (a) Accuracy and (b) MF1 for OSA detection and sleep staging.

false alarm rates during wakefulness and deep sleep. Fig. 7(d) shows a subject with severe OSA, where the model accurately captures a high density of respiratory events during N1 and N2, and demonstrates strong temporal alignment and stability in REM and N1 stages.

These visual results not only confirm the classification accuracy of MT-TASPPNet across various sleep states but also highlight its clinical potential and practicality in modeling complex temporal physiological events through its multi-task collaborative framework. Such individual-level insights help reveal not only the model's prediction consistency but also its potential to support clinical decision-making by offering transparent, temporally aligned outputs that mirror real physiological coupling patterns.

C. Advantages and Limitations

The proposed MT-TASPPNet demonstrates strong potential for multi-task sleep analysis with three main advantages.

First, it jointly models OSA detection and sleep staging within a unified architecture, leveraging modality-specific extractors and shared representations to capture physiological interactions and boost performance. Second, validation on a large real-world cohort (RNS, 778 subjects) confirms its robustness under heterogeneous annotations and signal quality, highlighting its clinical applicability. Third, the model maintains consistent performance across demographics and OSA severity, and excels in identifying transitional stages such as REM and N1, supporting its use in personalized sleep monitoring.

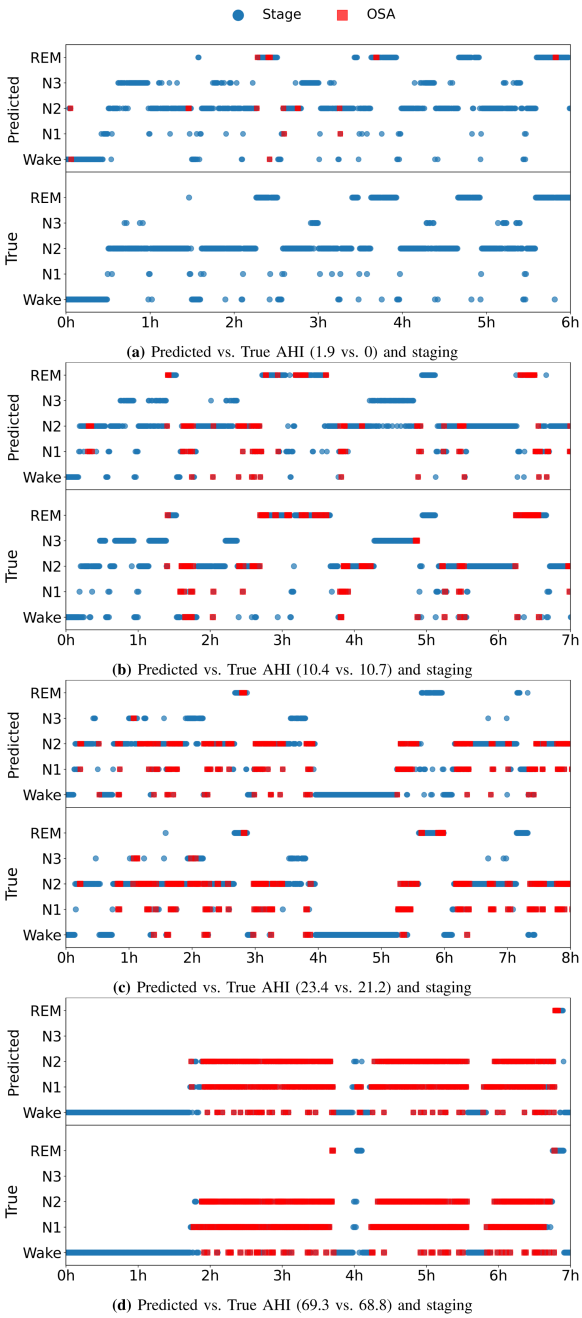
Although MT-TASPPNet demonstrates strong multi-task capabilities and performs well across multiple datasets, several limitations remain. First, its reliance on multi-channel PSG inputs and complex architecture leads to high computational demands, limiting its suitability for wearable or real-time applications. Multi-task training may also introduce early-stage gradient interference, affecting convergence.

Second, the model's generalization across centers still needs improvement. Cross-dataset validation revealed noticeable performance degradation on cohorts with differing demographics, acquisition protocols, and signal quality, highlighting the challenge of broader deployment.

Third, the model does not differentiate between types of apnea. For example, central and mixed apneas are treated as OSA, which may limit its applicability in more detailed clinical assessments.

Future research may address these challenges from several directions. First, model compression and architectural refinement can support lightweight and efficient deployment in practical settings. Second, robustness under limited-channel or single-modality inputs may be improved through methods such as modality completion, attention-based fusion, and contrastive learning. Third, to enhance generalizability across cohorts and recording environments, domain adaptation, invariant feature learning, and data harmonization techniques can be explored. Incorporating standardized evaluation frameworks, such as SleepyLand [43], may further improve the fairness and comparability of model assessment across diverse datasets. Furthermore, we acknowledge the lack of comprehensive benchmarking against state-of-the-art methods on both SHHS1 and SHHS2 datasets. This remains a limitation of our current work. Future efforts will aim to expand comparative evaluations and focus on enhancing the model's sleep staging performance on SHHS to better align with recent advances. Lastly, clinical interpretability and diagnostic precision may benefit from stratified detection

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(a) Predicted vs. True AHI (1.9 vs. 0) and staging

(b) Predicted vs. True AHI (10.4 vs. 10.7) and staging

(c) Predicted vs. True AHI (23.4 vs. 21.2) and staging

(d) Predicted vs. True AHI (69.3 vs. 68.8) and staging

Fig. 7. Visualization of predicted and true sleep staging with OSA events: (a) Healthy subject; (b) Mild OSA; (c) Moderate OSA; (d) Severe OSA. Each subfigure presents temporal alignment between predicted and true AHI and stage-specific apnea occurrences.

of apnea subtypes and severity levels, moving toward direct event detection and sleep-cycle-aware modeling to improve AHI estimation and clinical utility.

VII. Conclusion

In this study, we proposed a multi-task deep learning framework MT-TASPPNet for simultaneous OSA detection and sleep staging. The model employs ASPP modules in both modality-specific and shared layers to capture multi-scale features from heterogeneous physiological signals such as EEG, ECG, and airflow.

To support MTL, task-specific decoupled branches enable cross-task collaboration while minimizing interference. An EOG-guided prior further improves the classification of subtle stages like N1 and REM. Experiments on three datasets demonstrate strong performance, robustness under data heterogeneity, and good generalizability. MT-TASPPNet achieves competitive accuracy and F1 scores on both tasks and maintains consistency at the individual level. It shows the potential of joint modeling for sleep analysis and offers a scalable solution suitable for clinical and portable monitoring applications.

Acknowledgment

The authors gratefully acknowledge the contributions of the research staff and clinical teams involved in the collection and management of the Sydney Sleep Biobank data. The authors also extend our sincere thanks to all participants whose involvement made this research possible.

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